

현대광학(이.공) (PHY401-01)

학기	2026년 1학기
담당교수	이기준
이수구분	교선평수
학점	3.0
강의시간	화10:30-12:00(E7 - L21), 목10:30-12:00(E7 - L21)
강의실	E3 411 (이기준 교수실)
소속	기초학부

교과개요

With a prerequisite of <Electrodynamics> (2nd yr, 3 credits), this course will deal with different aspects of light or electromagnetic wave. Vectorial nature of light (polarization), wave nature of light (diffraction, interference) and quantum nature of light (photon, superposition, entanglement) will be covered.

교과목표

To let students have opportunity to contemplate upon what light really is, and how light interacts with matter. To let them appreciate the mathematical equivalence between spin-1/2 system and polarization state described in Jones vector. To introduce them into the world of nonlinear optics and quantum optics.

학습윤리

Just focus on how nature works, and ask a lot of questions during class with curious mind. Collaborations are encouraged among students, meaning I want students to help each other in climbing the mountain of Understanding of Light. Instructor will be fair in grading homeworks and exams, and students are expected to be fair as well when solving exam questions.

교과정책(방침)

Students are expected to practice the code of honors for all their homework and exams, although they are encouraged to collaborate elsewhere.